

ML in Applications

***Dipartimento di Automatica e Informatica
Politecnico di Torino, Torino, ITALY***

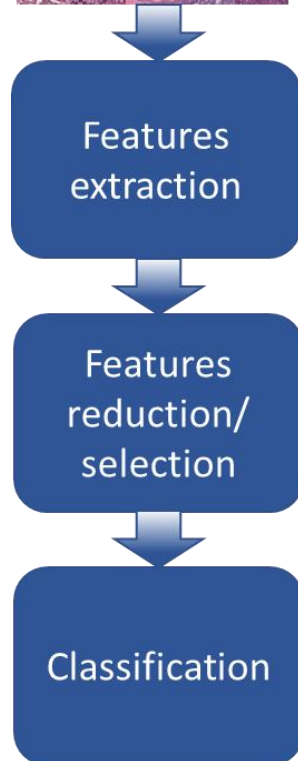
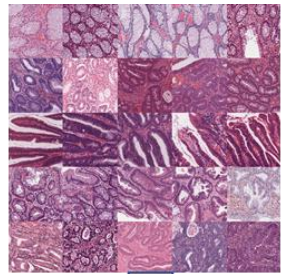


**Politecnico
di Torino**

Lab 5

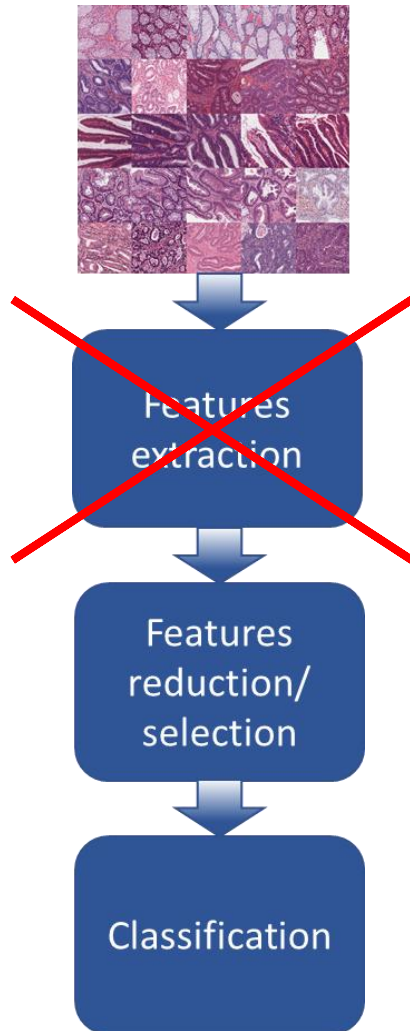
Transfer learning

Standard image analysis



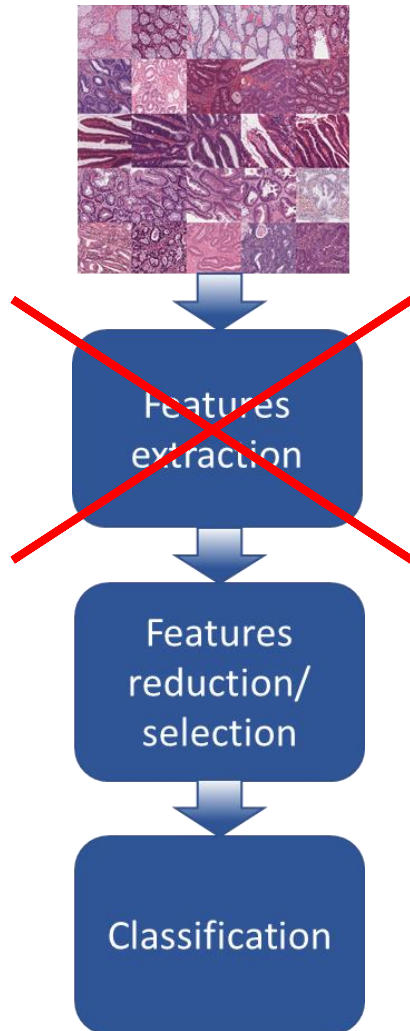
- Canonical machine-learning based images analysis:
 - Features extraction
 - Features reduction/selection
 - Classification
- The dependence on a **fixed set of handcrafted features** is a major limitation to the robustness

Why deep learning?



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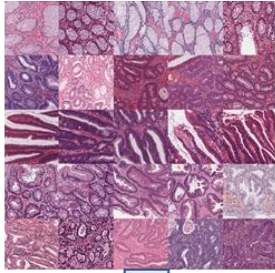
Why deep learning?



Pros:

- ✓ DL extract hierarchical knowledge from the data itself.
- ✓ No needs for handcrafted features

Why deep learning?



Features
extraction

Features
reduction/
selection

Classification

Cons:

- x Huge dataset (10^6 images)
- x High computational power
- x Difficult to find in everyday clinic.

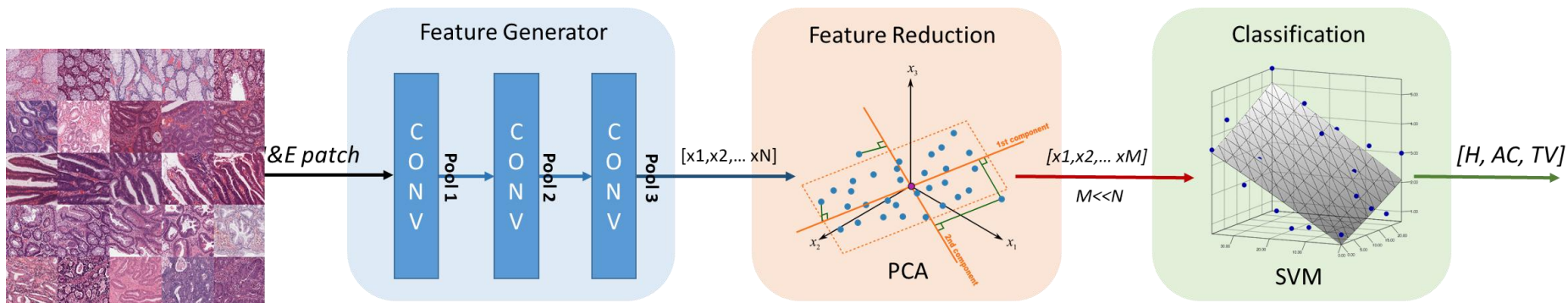
Transfer learning

- The **pretrained** model are trained on the ImageNet 2012 dataset (1.2 million photographs from 1000 different categories of natural objects).

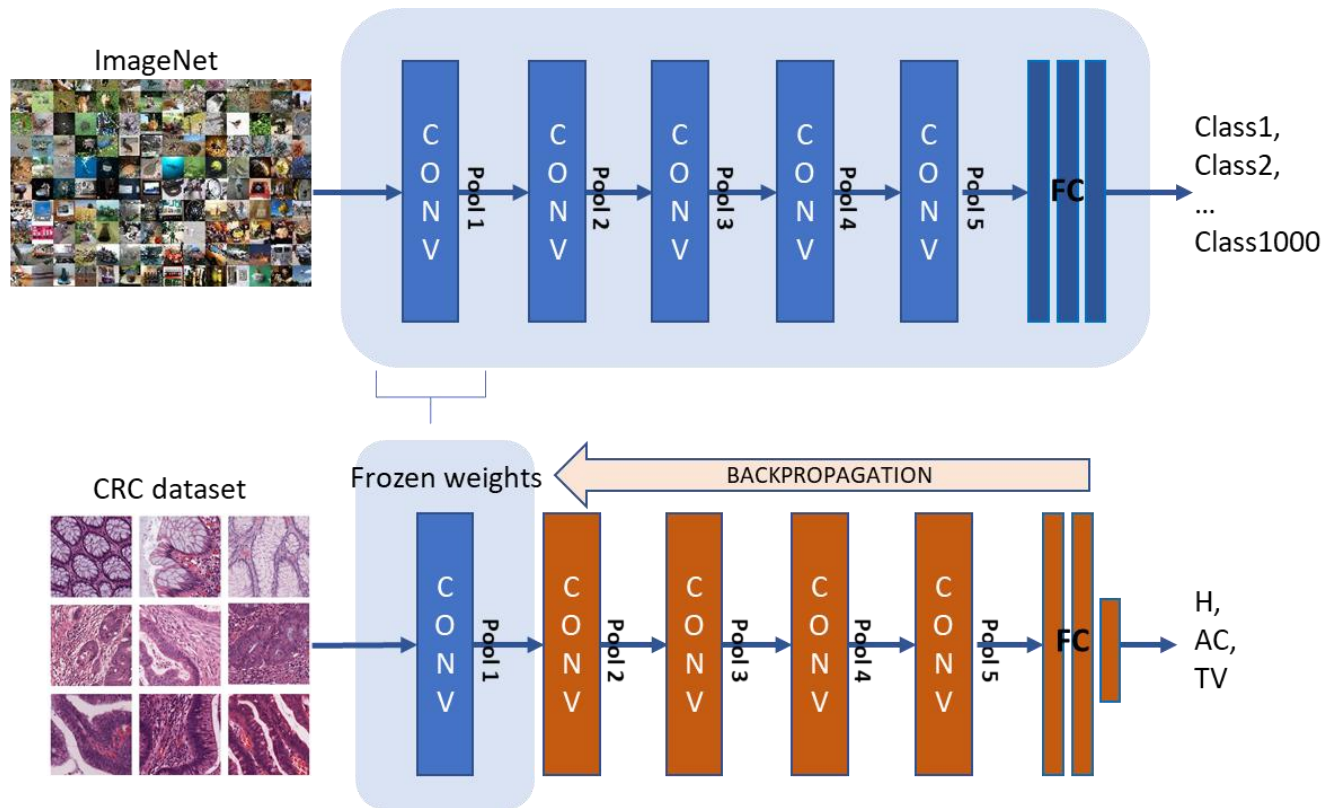


- Content and characteristics are completely different from our target.

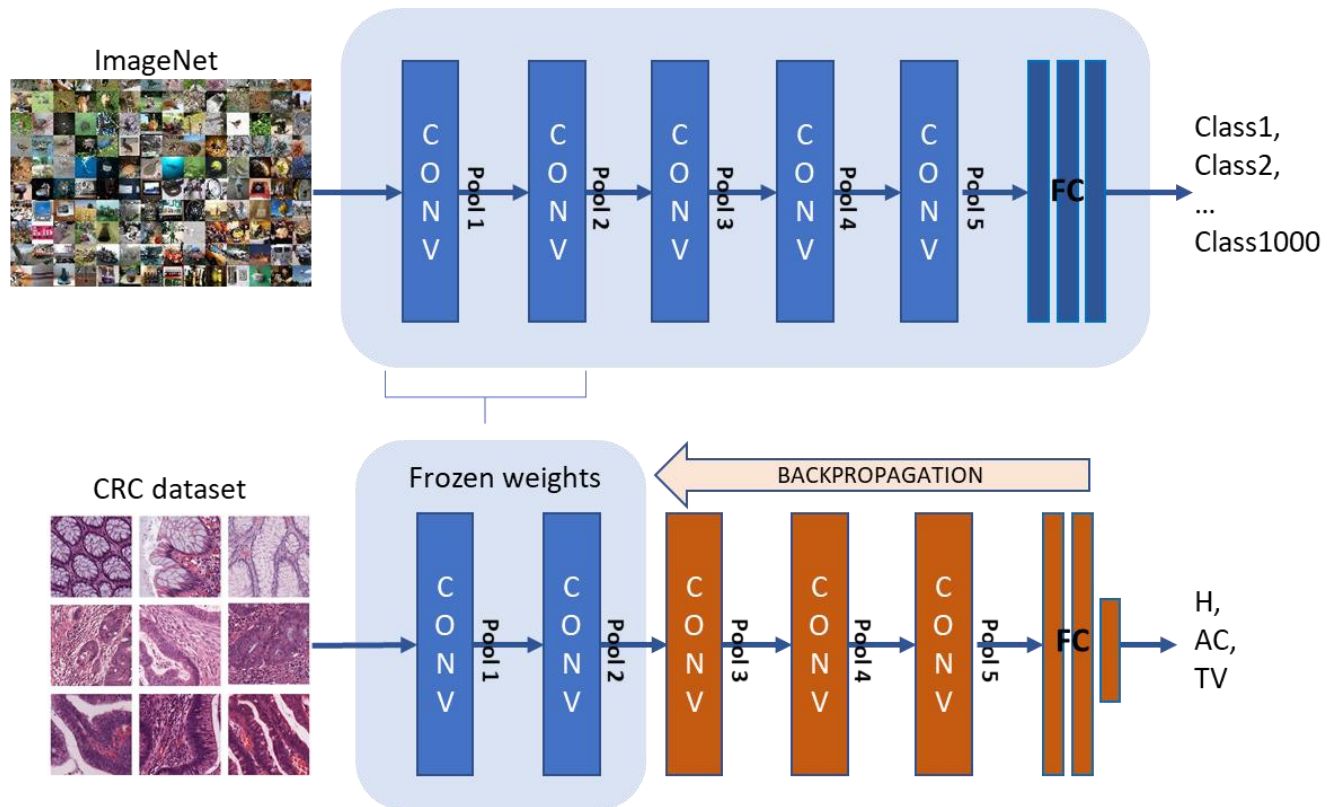
Features extraction



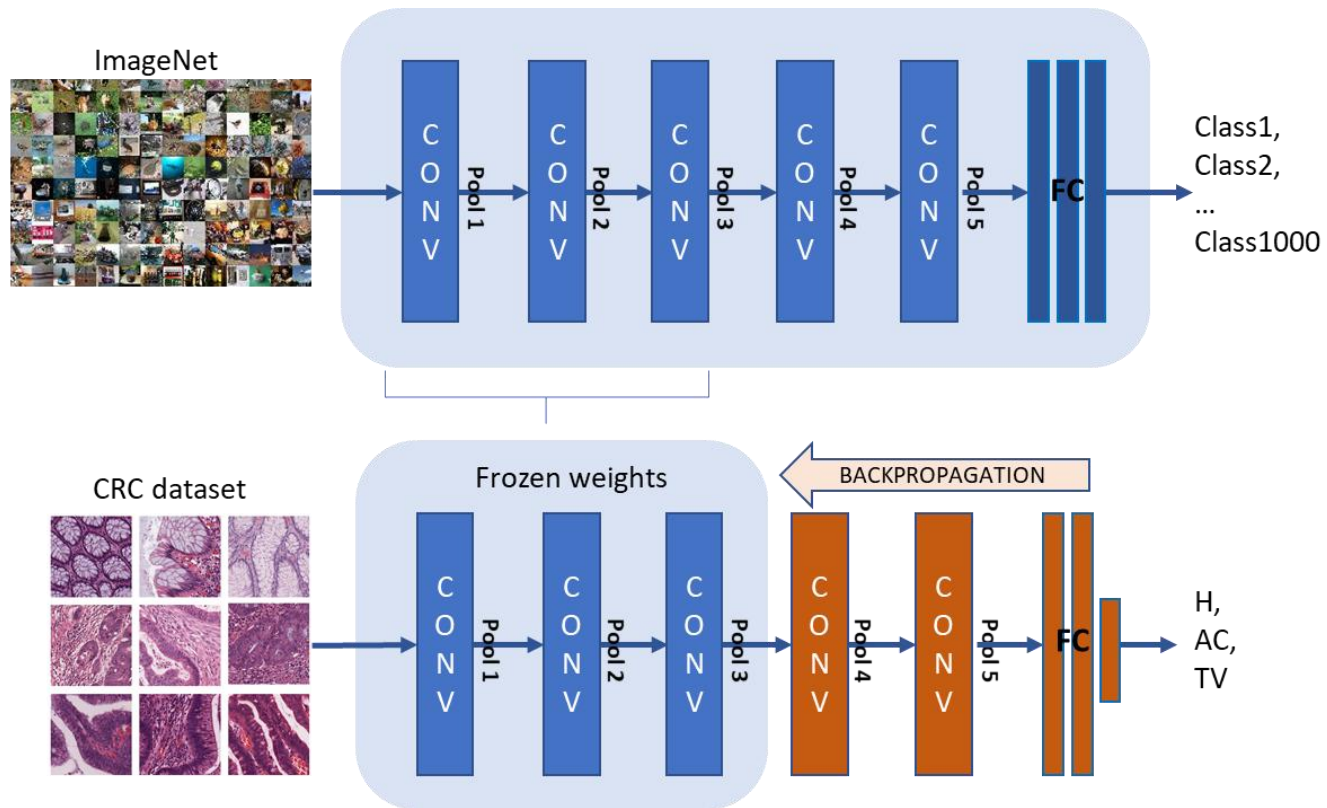
Fine tuning



Fine tuning



Fine tuning



Fine tuning

